Integrative Concept Analysis

Introduction

SilicaLayer Technologies is a global semiconductor equipment company that recently migrated from IntegraPLM to NexPhase PLM as its primary product lifecycle management (PLM) platform. The migration brought restructured workflows, new roles, and responsibilities for cross-functional teams — engineers, supply chain analysts, program managers, and quality professionals. To support the transition, our instructional design team built a training ecosystem that included day-in-the-life (DITL) eLearning courses, work instruction documents, instructor-led training (ILT) sessions, and job aids.

I write this analysis as both the instructional designer who built parts of this program and as someone embedded in the organization who has observed how employees experience it. That dual perspective is my lens for this paper.

Part 1: Fifteen Concepts in Practice

1. Acquisition Metaphor

The NexPhase PLM training program is built almost entirely on the acquisition metaphor — knowledge is packaged into modules, delivered to employees, and verified through completion tracking and knowledge checks. When an employee's progress bar reaches 100%, the organization considers them trained and ready to work in the live system. In practice, I have seen engineers who completed every module struggle significantly the first time they opened NexPhase PLM on a real production day because the module environment and real-world work are very different contexts. The metric looks clean, but it measures completion rather than capability.

2. Participation Metaphor

The participation metaphor, learning by becoming part of a community through real doing, is almost entirely absent from the formal training program. There is no structured path for a new NexPhase PLM user to progress from beginner to confident practitioner alongside experienced colleagues. The ILT sessions offer a brief shared space, but the structure remains one-directional. Employees finish training with information about NexPhase PLM but still do not feel they belong to the Teamcenter-using community at SilicaLayer Technologies.

3. Human-Centered Design

Our team made a real effort toward human-centered design by choosing the DITL format, structuring content around what an employee actually does in a real workday, rather than around system features in a menu tour. However, the design process was still driven primarily by subject-matter experts and project timelines rather than by learners themselves. Feedback from pilot users revealed workflow inaccuracies that reflected how SMEs thought work happened, not how employees actually did it. A more fully human-centered process would have started with sustained observation of real work before any module was built.

4. Participatory Design

End-users, engineers, and supply chain coordinators who would live with the new workflows were consulted during review cycles, but were never part of the design team itself. Their role was to validate content that had already been built, not to shape what was built or how. Several engineers flagged during reviews that task sequences did not match how their teams actually managed engineering change orders, but many of those corrections were deferred due to timeline pressure. A genuinely participatory process would have caught those mismatches before development, not after.

5. Equity-by-Design

Our cheatsheets and IntegraPLM-versus-NexPhase PLM comparison job aids were a genuine equity move. They gave every employee a concrete reference point, regardless of prior system experience, and were especially helpful for team members who were entirely new to PLM tools or needed to reference steps during live work rather than relying on memory alone. The gap is that the core learning pathways were not differentiated by experience level, language background, or workload. As a result, employees navigating peak production demands, with limited prior experience with enterprise systems, or in non-English-dominant environments, had to work harder than others just to keep up with the same standardized content.

6. Third Space

Our IntegraPLM-versus-NexPhase PLM comparison job aids came closest to creating a third space, because they explicitly placed the old and new systems side by side and helped employees map their existing IntegraPLM knowledge onto the new workflows —

treating the transition as a translation rather than a complete replacement of what they knew. The modules themselves, however, rarely made this connection, presenting the NexPhase PLM way of doing things as complete and authoritative without acknowledging what employees already knew from years of working in IntegraPLM, leaving the deep process knowledge employees brought into training unused rather than activating it as a foundation for new learning.

7. Identity in Learning

We acknowledged role identity through our role-based DITL courses, so a supply chain analyst was not sitting through engineering change order workflows irrelevant to their job, which gave employees a sense that the training recognized who they were and what they actually did at SilicaLayer Technologies. What we did not address was experience-level identity — senior engineers who had managed PLM processes for over a decade moved through the same foundational content as employees who had never used a PLM system before, which quietly signaled that their expertise did not count and led some of them to disengage from sessions that treated them as novices.

8. Funds of Knowledge

Our DITL courses made a genuine effort to activate funds of knowledge by grounding scenarios in real SilicaLayer Technologies workflows rather than generic examples, so a supply chain scenario felt recognizable and contextually meaningful to someone who actually did that work, which helped employees connect what they already knew to what they were learning. What we could have done better is to explicitly invite that prior knowledge at the start of each module, something as direct as acknowledging what employees already knew about PLM before showing them how NexPhase PLM handles it, because several employees told me they felt like they were learning in a vacuum, disconnected from the years of work experience they brought into the training.

9. Discourse & Participation Structures

Our asynchronous sessions were a genuine attempt to open up the participation structure. Employees could ask questions, share observations, and work through the tool together, and some real back-and-forth did happen, where an employee's workflow question shifted the direction of the session in a meaningful way. The DITL modules themselves, however, remain entirely one-directional: the content speaks, and the learner clicks through, with no channel to contribute their own experience, raise a concern about how a new workflow affects their team, or push back on a process step that does not match their operational reality.

10. Turn-Taking Patterns

In our asynchronous sessions, turn-taking was meaningfully better than in typical ILT sessions. Employees asked genuine questions, occasionally pushed back on process steps, and engaged in real exchanges in which both the facilitator and participants contributed knowledge, closer to the multi-directional participation pattern that the Sean Numbers classroom made visible. In the eLearning modules, the pattern is entirely one-sided: the module presents, the learner clicks Next, and the only learner's turn is a knowledge check with a predetermined correct answer, a very thin form of interaction that gives employees no real opportunity to bring their own thinking into the learning environment.

11. Multimodality

Our training was genuinely more multimodal than typical corporate eLearning: DITL simulations engaged visual and kinesthetic channels simultaneously as employees clicked through real NexPhase PLM interface screens; annotated screenshots in work instructions provided visual anchors for procedural steps; and sandbox access gave employees a real environment to explore physically on their own terms. The remaining gap is that sandbox exploration was unstructured and the simulations followed a scripted path, so employees never experienced the embodied challenge of navigating a complex system under real conditions — the kind of hands-on, consequential engagement that the Embodying STEM readings identify as foundational for genuine technical competence.

12. Lifelong & Lifewide Learning

Our job aids and cheat sheets recognized the lifelong dimension of learning in a small but meaningful way, because they were designed as ongoing reference tools rather than one-time training artifacts, acknowledging that employees would continue learning NexPhase PLM long after the formal modules ended and would need resources that worked at the point of real need. What we did not design for is the diversity of prior learning lives each employee brought into this training: some had used SAP or other PLM systems at previous companies, some had deep self-taught knowledge of SilicaLayer Technologies's processes, and some were brand new to enterprise systems entirely, yet all moved through identical content with no diagnostic, no differentiated path, and no acknowledgment that their learning histories existed.

13. Interest-Driven Learning

Our DITL courses were a real step toward interest-driven design; a supply chain analyst's course was built around the workflows they owned, not a generic system demo. This made the content feel relevant to their job and gave employees a reason to engage beyond compliance. The gap is that we connect training to roles, but not always to the specific goals that motivate people in those roles. An engineer's real interest is getting a change order approved without holding up the production line. Making that the explicit frame for every relevant module would have sharpened motivation considerably and made the connection between learning and real work much more direct.

14. Informal Learning Environments

The most active learning ecosystem around the NexPhase PLM migration has been informal and largely self-organized, employees built their own Teams channels for questions, early-confident users became unofficial go-to resources for their teams, and some groups created their own quick-reference guides that circulated more widely than the official job aids, all of which represents genuine community-driven learning that the formal program did not design for but that emerged powerfully around it. This informal ecosystem is a real strength of how the migration unfolded at SilicaLayer Technologies, and designing the formal program to acknowledge, connect to, and amplify this informal learning, rather than treating it as the complete solution, would make the overall training strategy significantly more effective.

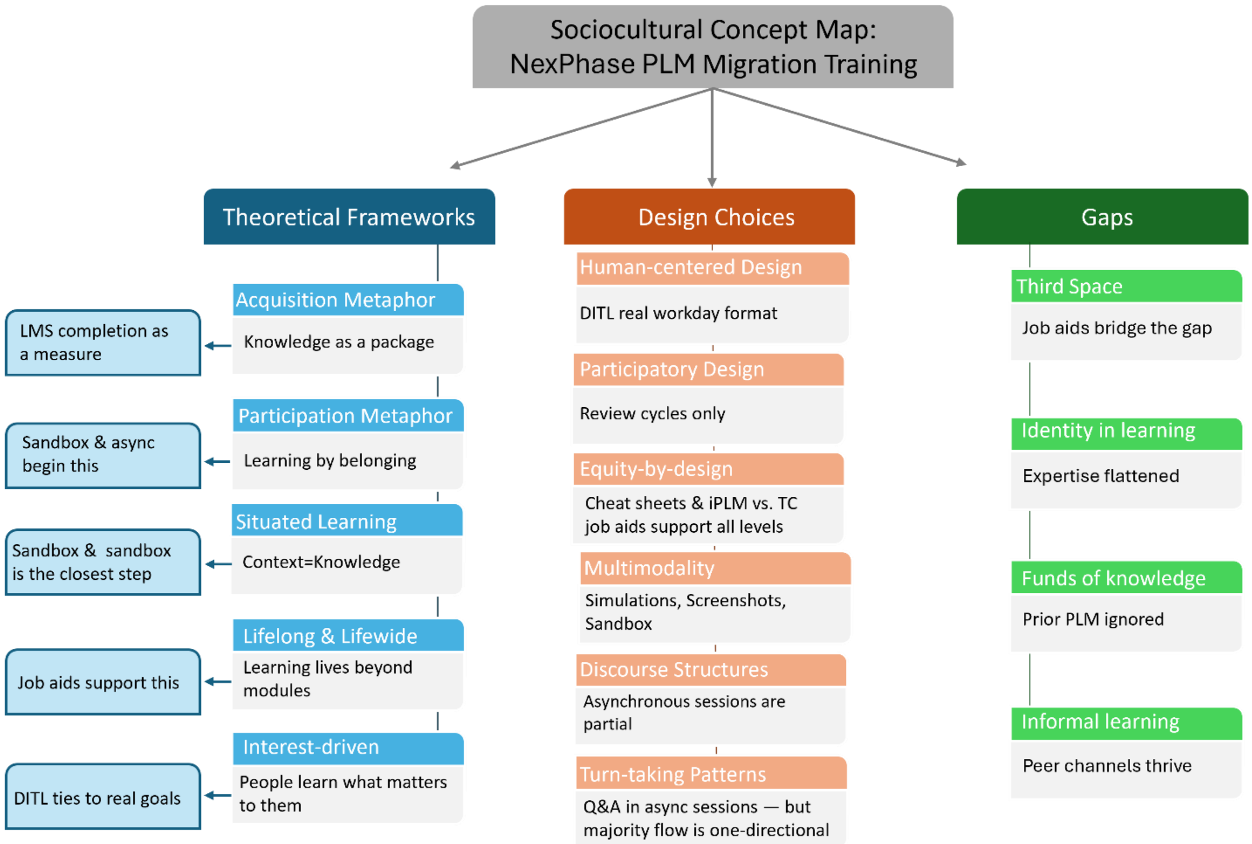
15. Situated Learning

Our sandbox access was the closest the formal program came to honoring situated learning theory, because giving employees a real NexPhase PLM environment to explore on their own —

even without live production data — created a learning context meaningfully closer to actual work than a guided simulation, and employees who used it actively built the embodied system familiarity that modules alone cannot produce. The remaining gap is that even the sandbox is decontextualized from real work — there are no real stakes, no real colleagues, and no real decisions with production consequences — so the employees who became most capable in NexPhase PLM were ultimately those who had early access to the live system alongside an experienced colleague doing real work, which is situated learning happening naturally in the spaces the formal program did not design for.

Part 2: Concept Map

The concept map below organizes all fifteen concepts around the central tension I noticed throughout this analysis: the gap between a training program built to deliver and verify knowledge and what it actually takes to become a confident, capable NexPhase PLM user in a real-world work environment. I grouped the concepts into three layers: the theoretical frames that explain how learning works, the design choices our team made and how well they hold up, and the things that are still missing or only just starting to emerge.



Reflection

Looking at this analysis as a whole, I feel genuinely proud of what our cross-cultural team from India and the USA built, and more clearly aware of where we can go further. The DITL courses, the IntegraPLM-versus-NexPhase PLM comparison materials, the sandbox access, and the asynchronous sessions were all deliberate choices rooted in what we believed our learners needed — and many of those choices show up positively across multiple sociocultural concepts in this analysis. That matters because it means the program was not built blindly on the acquisition metaphor alone — there were real participatory, multimodal, and human-centered instincts in the design, even when they were not framed in those theoretical terms.

At the same time, writing this paper has shown me where the acquisition model quietly reasserted itself even when we thought we were doing something more participatory. Giving sandbox access is excellent, but unguided sandbox exploration is not the same as situated practice within a real community of work. Involving users in review cycles is meaningful, but it is not the same as participatory design. Our job aids created a third space at the resource level, but the modules themselves never opened that space for the learner inside the primary learning experience.

The most important thing these fifteen concepts have given me is a more precise vocabulary for design decisions I will make going forward. Questions like whose knowledge we are activating before we start, how close this learning is to real work, and whom this design leaves behind will shape how I approach projects like this from the beginning — not as a critique of what was built, but as a framework for building something better next time.

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